Summation based on hyperbolic Series.

In hyperbolic series, C+iS method is not applicable so to sum the series of hyperbolic sines or cosines we follow the following steps.

1. Replace $\sinh x$ by $\frac{e^{x}-e^{-x}}{2}$ and $\cosh x$ by $\frac{e^{x}+e^{-x}}{2}$.
2. Represent the series in e^{x} and e^{-x}
3. Sum up each series using standard results
4. Put result in terms of hyperbolic sines or cosines.

Q. $\cosh\alpha - \frac{1}{2}\cosh 2\alpha + \frac{1}{3}\cosh 3\alpha - \dots$

Replace $\cosh\alpha = \frac{e^{\alpha}+e^{-\alpha}}{2}$

$$\Rightarrow \frac{e^{\alpha}+e^{-\alpha}}{2} - \frac{1}{2}\left(\frac{e^{2\alpha}+e^{-2\alpha}}{2}\right) + \frac{1}{3}\left(\frac{e^{3\alpha}+e^{-3\alpha}}{2}\right) - \dots$$

$$= \frac{1}{2}\left[e^{\alpha} - \frac{e^{2\alpha}}{2} + \frac{1}{3}e^{3\alpha} - \dots\right] + \frac{1}{2}\left[e^{-\alpha} - \frac{1}{2}e^{-2\alpha} + \frac{1}{3}e^{-3\alpha} - \dots\right]$$

$$= \frac{1}{2}\left[\log(1+e^{\alpha}) + \log(1+e^{-\alpha})\right]$$

$$= \frac{1}{2}\left[\log\left[(1+e^{\alpha})(1+e^{-\alpha})\right]\right]$$

$$= \frac{1}{2}\log\left(1+e^{\alpha}+e^{-\alpha}+e^{\alpha}\cdot e^{-\alpha}\right)$$

$$= \frac{1}{2}\log\left(2+e^{\alpha}+e^{-\alpha}\right)$$

$$= \frac{1}{2}\log\left(2+\frac{e^{\alpha}+e^{-\alpha}}{2}\cdot 2\right)$$

$$= \frac{1}{2}\log\left(2+2\cosh\alpha\right)$$

$$= \frac{1}{2}\log 2\left(1+\cosh\alpha\right) \qquad 1+\cosh 2\alpha = 2\cosh^2\alpha$$

$$= \frac{1}{2}\log\left(2\cdot 2\cosh^2\alpha/2\right)$$

$$= \frac{1}{2}\log\left(4\cosh^2\alpha/2\right)$$

$$= \frac{1}{2}\log\left(2\cosh\alpha/2\right)^2$$

$$= \frac{1}{2}\cdot 2\log\left(2\cosh\alpha/2\right) \qquad \left(\log a^b = b\log a\right)$$

$$= \log\left(2\cosh\alpha/2\right)$$

Q. $\sinh\alpha - \frac{1}{2}\sinh 2\alpha + \frac{1}{3}\sinh 3\alpha - \dots$

Replace $\sinh\alpha = \frac{e^{\alpha}-e^{-\alpha}}{2}$

$$\frac{e^{\alpha}-e^{-\alpha}}{2} - \frac{1}{2}\left(\frac{e^{2\alpha}-e^{-2\alpha}}{2}\right) + \frac{1}{3}\left(\frac{e^{3\alpha}-e^{-3\alpha}}{2}\right) - \dots$$

$$= \frac{1}{2}\left[\left(e^{\alpha} - \frac{1}{2}e^{2\alpha} + \frac{1}{3}e^{3\alpha} - \dots\right) - \left(e^{-\alpha} - \frac{1}{2}e^{-2\alpha} + \frac{1}{3}e^{-3\alpha} - \dots\right)\right]$$

$$= \frac{1}{2}\left[\log(1+e^{\alpha}) - \log(1+e^{-\alpha})\right]$$

$$= \frac{1}{2}\log\left(\frac{1+e^{\alpha}}{1+e^{-\alpha}}\right)$$

$$= \frac{1}{2}\log\left(\frac{1+e^{\alpha}}{1+1/e^{\alpha}}\right)$$

$$= \frac{1}{2}\log\left[\frac{e^{\alpha}(1+e^{\alpha})}{(e^{\alpha}+1)}\right]$$

$$= \frac{1}{2}\log e^{\alpha} = \frac{1}{2}\cdot\alpha$$

Q. $x\sinh x + x^2 \sinh 2x + x^3 \sinh 3\alpha + \cdots$

Replace $\sinh\alpha = \dfrac{e^{\alpha}-e^{-\alpha}}{2}$

$$x\left(\frac{e^{\alpha}-e^{-\alpha}}{2}\right) + x^2\left(\frac{e^{2\alpha}-e^{-2\alpha}}{2}\right) + x^3\left(\frac{e^{3\alpha}-e^{-3\alpha}}{2}\right) + \cdots$$

$$= \frac{1}{2}\left[xe^{\alpha} + x^2e^{2\alpha} + x^3e^{3\alpha} + \cdots\right] - \frac{1}{2}\left[xe^{-\alpha} + x^2e^{-2\alpha} + x^3e^{-3\alpha} + \cdots\right]$$

$$= \frac{1}{2}\times\frac{xe^{\alpha}}{1-xe^{\alpha}} - \frac{1}{2}\cdot\frac{xe^{-\alpha}}{1-xe^{-\alpha}}$$

$$= \frac{1}{2}\left[\frac{xe^{\alpha} - x^2\cancel{e^{\alpha}}\cdot e^{-\alpha} - xe^{-\alpha} + x^2\cancel{e^{-\alpha}}\cdot e^{\alpha}}{1-xe^{-\alpha}-xe^{\alpha}+x^2e^{\alpha}\cdot e^{-\alpha}}\right]$$

$$= \frac{1}{2} \cdot \frac{x(e^{\alpha} - e^{-\alpha})}{1 + x^2 - x(e^{\alpha} + e^{-\alpha})}$$

$$= \frac{1}{\cancel{2}} \cdot \frac{x \times \cancel{2} \sinh\alpha}{1 + x^2 - 2x\cosh\alpha}$$

$$= \frac{x\sinh\alpha}{1 - 2x\cosh\alpha + x^2}$$